

Answer all questions.

1. Road traffic accidents occur when a vehicle is unable to stop safely. The overall stopping distance can be worked out using the following equation:

overall stopping distance = thinking distance + braking distance

The table shows stopping distances from the Highway Code.

Speed (mph)	20	30	40	50	60	70
Thinking distance (m)	6	9	12	15		21
Braking distance (m)	6	14	24	38	56	75
Overall stopping distance (m)	12	23	36	53		96

- (a) Complete the table. [2]

- (b) (i) Describe how worn tyres affect the following distances. [2]

Thinking distance

Braking distance

- (ii) Describe how a driver using a mobile phone affects the following distances. [2]

Thinking distance

Braking distance

6

